

stepper motors



gear box
brake
encoder
up to IP68
special versions



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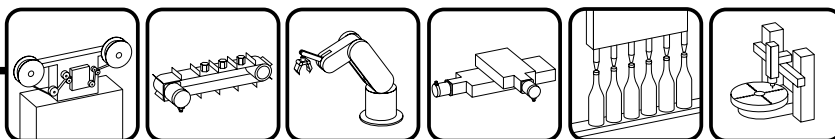
stögra
ANTRIEBSTECHNIK GMBH



○ **driving**

○ **motion control**

○ **positioning**



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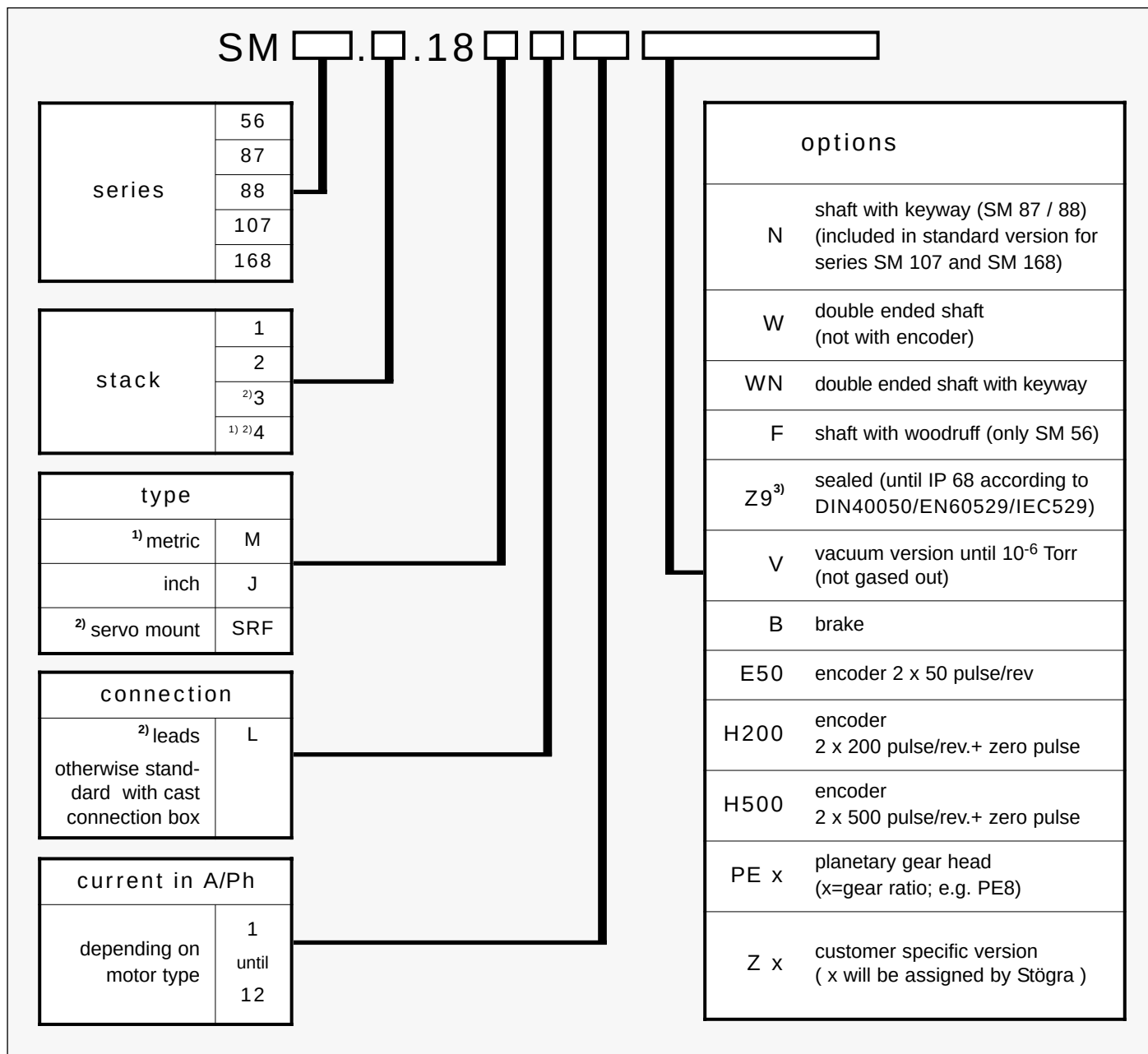
Mechanical and electrical ratings and dimensions are, therefore, subject to change without notice.

No liability whatsoever is accepted.

ordering key	4
dimensions standard type	5
keyways and woodruffs	6
overview electrical and mechanical specifications	7
max. operating torque	
series SM 56	8
series SM 87	10
series SM 88	12
series SM 107	14
series SM 168	16
stepping motor phase current characteristics	16
hollow shaft motors	
series SMH 88	17
series SMH 107	17
standard stepping motor connections SM 56 – SM 168	18
table stepping motors – controls / drives	19
stepping motor equipment	
overview: stepping motor with gear, encoder and brake	20
stepping motor series SM 56 with D-Sub connector	20
stepping motor with planetary gear head series SM 56 PE	21
stepping motor with planetary gear head series SM 87 PE / SM 88 PE	22
stepping motor with planetary gear head series SM 107 PE	23
stepping motor with integrated encoder E50	24
stepping motor with integrated encoder H200 and H500	25
stepping motor with brake	26
dimensions stepping motor with gear, encoder and brake	27
cable glands	27

4 ordering Key

STÖGRA Stepper motors are designed as modular system. This enables us to provide a great variety of standard motor types and a high flexibility for customer specific solutions.



¹⁾ not for series 56

²⁾ not for series 168

³⁾ IP68 according to DIN 40050 / EN60529 / IEC529 – IP58 according to VDE0530-5 / EN60034-5 / IEC34-5

Please note, not all options can be combined!

SM 86 series is replaced by SM 87 series, SM 108 and SM 109 series is replaced by SM 107 series.

Cable glands are not included in standard motor deliveries! Cable glands must be ordered separately.

general STÖGRA motor specifications:

- ± 3% accuracy based on 1.8° motor step angle (non cumulative)
- operating temperature -30°C until 80°C (short time until 100°C) for standard types
- insulation class F according to VDE 0530
- dielectric motor strength 1800 vrms (series 56: 1000 vrms)
- high bearing thrust and overhang loads

order examples:

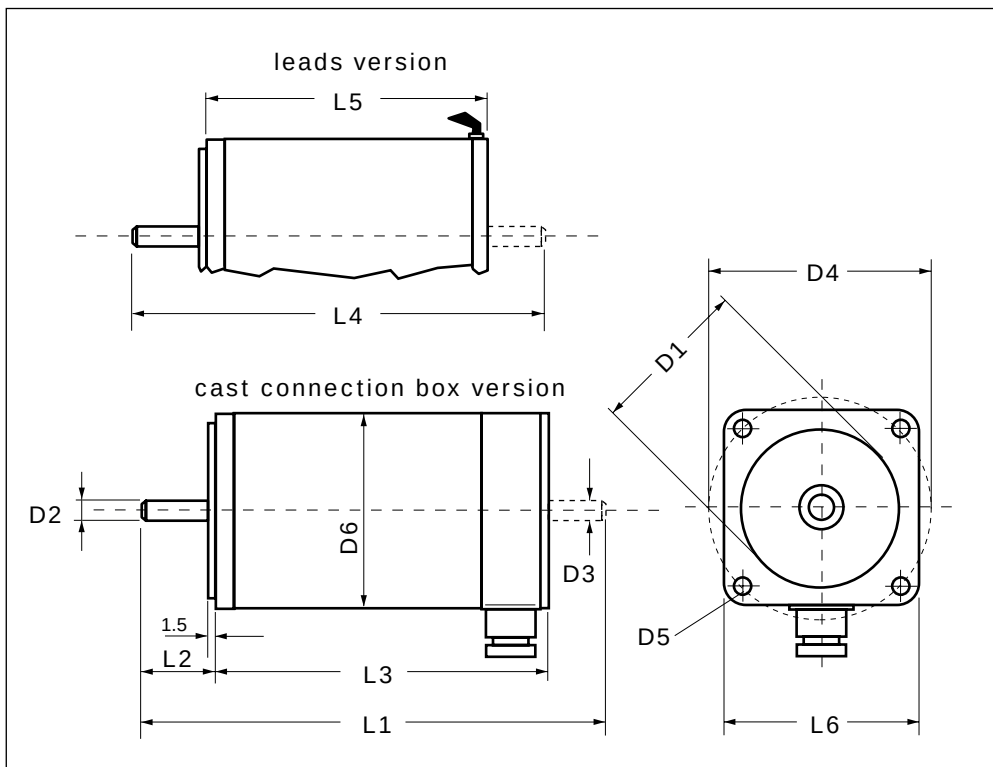
SM 56.2.18J3 E50 PE8

SM 87.1.18ML3

SM 88.3.18M8 BH200

SM 107.2.18M12 BE50 PE4

SM 168.2.18M12



stepping motor		D1 -0.05		D2 -0.02		D3 -0.02		D4		D5	D6	L1	L2	L3	L4	L5	L6	thread of cable entry					
series	type	M	J	M	J	M	J	M	J				+0.5	±0.5		±0.5							
56	SM 56.1.18		38.1		6.35		6.35		66.5	5.3	56.5	108	21	76	90	50	56.5	M20 x 1.5					
	SM 56.2.18											134		102	116	76							
	SM 56.3.18											162		130	144	104							
87	SM 87.1.18	73		10 (12) ¹⁾	9.52	10 (12) ¹⁾	9.52	99		6.5	86	137	31.5	85.5	137	60.5	86	M20 x 1.5					
	SM 87.2.18											169		117.5	169	92.5							
	SM 87.3.18											201		149.5	201	124.5							
	SM 87.4.18											233		181.5	233	156.5							
88	SM 88.1.18	73		12 (10) ¹⁾	9.52	12 (10) ¹⁾	9.52	99		6.5	86	145	31.5	93.5	145	68.5	86	M20 x 1.5					
	SM 88.2.18											177		125.5	177	100.5							
	SM 88.3.18											209		157.5	209	132.5							
	SM 88.4.18											241		189.5	241	164.5							
107	SM 107.1.18	60	55.54	12	12.7	10	12.7	127.5	125.5	8.5	108	170	32	111		89.5	108	M20 x 1.5					
	SM 107.2.18			16	15.87	12						238	50	161		139.5							
	SM 107.3.18											288		211		189.5							
	SM 107.4.18											338		261		239.5							
168	SM 168.1.18	180		24		19		215		15	168	268	50.5	179			192	M20 x 1.5					
	SM 168.2.18											343		254									

¹⁾ series SM87 also available with 12 mm shaft and series SM88 also available with 10 mm shaft

all dimensions in mm

M = metric

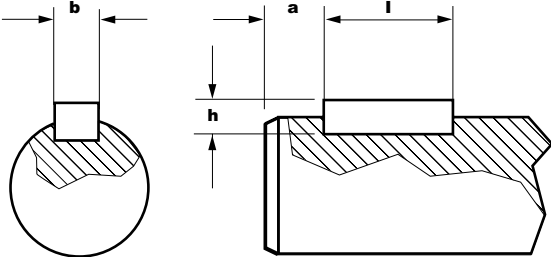
J = inch

6 keyways and woodruffs

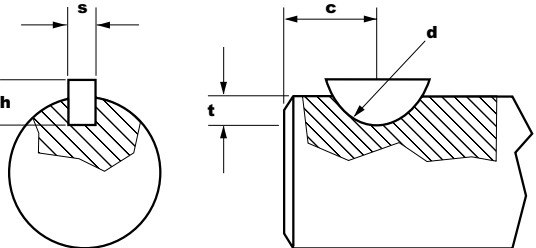
keyways:

Series SM 107 and SM 168 are delivered in standard version with keyway (but not on double ended shaft).

At series SM 87 the keyway is an option, which has to be indicated at the ordering key.

keyway DIN 6885 T1		stepping motor				keyway DIN 6885 T1		
		series	type	M	J	SRF	type A b x h x l	¹⁾ a
							a	
		56	SM 56.1.18	/	0	0	A2 x 2 x 12	3
			SM 56.2.18					
			SM 56.3.18					
		87	SM 87.1.18	0	0	0	until Ø10 A3 x 3 x 15 from Ø12 A4 x 4 x 15	6
			SM 87.2.18					
			SM 87.3.18					
			SM 87.4.18					
		88	SM 88.1.18	0	0	0	until Ø10 A3 x 3 x 15 from Ø12 A4 x 4 x 15	6
			SM 88.2.18					
			SM 88.3.18					
			SM 88.4.18					
		107	SM 107.1.18	S	/	/	A5 x 5 x 20	5
			SM 107.2.18					
			SM 107.3.18					
			SM 107.4.18					
		168	SM 168.1.18	S	/	/	A8 x 7 x 25	5
			SM 168.2.18					

¹⁾ double ended shaft

woodruff DIN 6888		stepping motor				woodruff DIN 6888			
		series	type	J	SRF	type s x h	d	c	t
		56	SM 56.1.18	0	0	2 x 2.6	7	7	1.8
			SM 56.2.18						
			SM 56.3.18						

/ = no standard stepping motor

S = standard – stepping motor is delivered with keyway in standard version

O = option – stepping motor is delivered without keyway or woodruff in standard version

order examples:

SM 56.2.18J3F	with woodruff
SM 87.2.18M6N	with keyway at front shaft
SM 87.2.18M6WN	with keyway only at double ended shaft
SM 87.2.18M6NWN	with keyway at front and double ended shaft

overview electrical and mechanical specifications 7

electrical and mechanical specifications

weight and rotor inertia are for standard versions with cast connection box, without double ended shaft		electrical specifications				mechanical specifications						
		resistance per phase	inductance per phase	current per phase unipolar	current per phase bipolar	step angle (at full step)	holding torque	detent torque	rotor inertia	bearing thrust load	bearing overhang load	weight
series	motor type	Ohm	mH	A	A	°	Nm	Nm	kgcm ²	N	N	kg
56	SM 56.1.18 J1	4.75	9	1	1.4	1.8	0.45	0.01	0.125	80	150	0.6
	SM 56.1.18 J3	0.72	1	3	4.2							
	SM 56.1.18 J3.9	0.42	0.64	3.9	5.5							
	SM 56.2.18 J1.5	3.9	9	1.5	2.1	1.8	0.85	0.017	0.25	80	150	1
	SM 56.2.18 J2	2.6	5	2	2.8							
	SM 56.2.18 J3	1.2	2.6	3	4.2							
	SM 56.3.18 J1.5	4.3	9	1.5	2.1	1.8	1.25	0.025	0.375	80	150	1.35
	SM 56.3.18 J3	1.46	3	3	4.2							
	SM 56.3.18 J4.6	0.72	1.2	4.6	6.5							
87	SM 87.1.18 M1.6	2.9	6	1.6	2.3	1.8	1.8	0.026	0.65	180	280	1.7
	SM 87.1.18 M3	0.72	1.6	3	4.2							
	SM 87.1.18 M5	0.28	0.7	5	7							
	SM 87.2.18 M3.5	0.74	3	3.5	5	1.8	3.6	0.05	1.3	180	280	2.65
	SM 87.2.18 M4.6	0.48	1.5	4.6	6.5							
	SM 87.2.18 M6	0.38	1	6	8.4							
	SM 87.3.18 M3.5	1.1	5	3.5	5	1.8	5.4	0.08	1.95	180	280	3.65
	SM 87.3.18 M6	0.43	1.7	6	8.4							
	SM 87.3.18 M7	0.33	1	7	10							
	SM 87.4.18 M6	0.55	2.3	6	8.4	1.8	7.2	0.1	2.6	180	280	4.6
	SM 87.4.18 M7	0.42	1.8	7	10							
88 ¹⁾	SM 88.1.18 M2	1.88	11.1	–	2	1.8	3	0.042	1.35	180	280	1.9
	SM 88.1.18 M4	0.5	2.5	–	4							
	SM 88.1.18 M8	0.13	0.75	–	8							
	SM 88.2.18 M2	3.61	26	–	2	1.8	6	0.08	2.7	180	280	2.85
	SM 88.2.18 M4	0.74	5.5	–	4							
	SM 88.2.18 M8	0.21	1.5	–	8							
	SM 88.3.18 M4	1.14	10.9	–	4	1.8	9	0.13	4.05	180	280	3.85
	SM 88.3.18 M8	0.29	2.6	–	8							
	SM 88.3.18 M12	0.14	1	–	12							
	SM 88.4.18 M8	0.37	3.55	–	8	1.8	12	0.16	5.4	180	280	4.8
	SM 88.4.18 M12	0.12	1.75	–	12							
107	SM 107.1.18 M6	0.3	1.6	5	7	1.8	5	0.11	4	400	650	4.3
	SM 107.1.18 M8	0.225	1.2	5.7	8							
	SM 107.1.18 M12	0.1	0.55	8.8	12.5							
	SM 107.2.18 M8	0.38	2.4	5.7	8	1.8	9	0.21	8	400	650	7.2
	SM 107.2.18 M10	0.25	1.6	7.1	10							
	SM 107.2.18 M12	0.175	1.15	8.8	12.5							
	SM 107.3.18 M10	0.38	2.7	7.1	10	1.8	13	0.3	12	400	650	9.8
	SM 107.3.18 M12	0.28	1.9	8.8	12.5							
	SM 107.4.18 M12	0.34	2.7	8.8	12.5	1.8	17	0.4	16	400	650	12.5
168	SM 168.1.18 M12	0.18	2.5	8.8	12.5	1.8	19	0.3	31.2	660	1000	18
	SM 168.2.18 M12	0.28	5	8.8	12.5	1.8	38	0.6	64.4	660	1000	23

¹⁾ SM88 only with bipolar winding in standard version

8 max. operating torque series SM 56

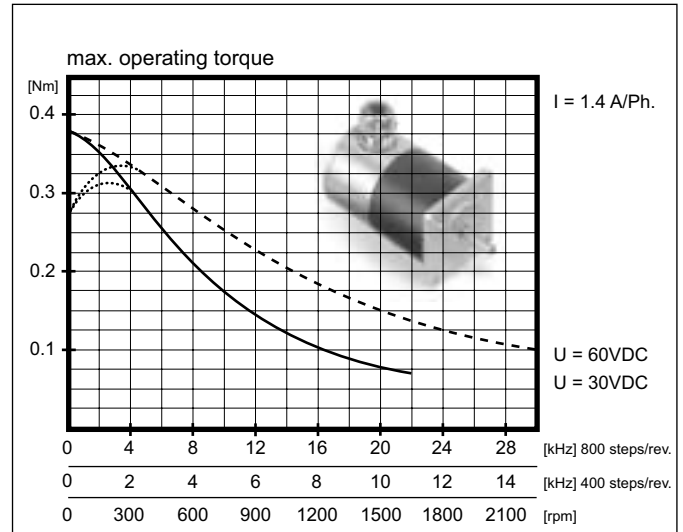


SM 168 SM 107 SM 87/88 SM 56

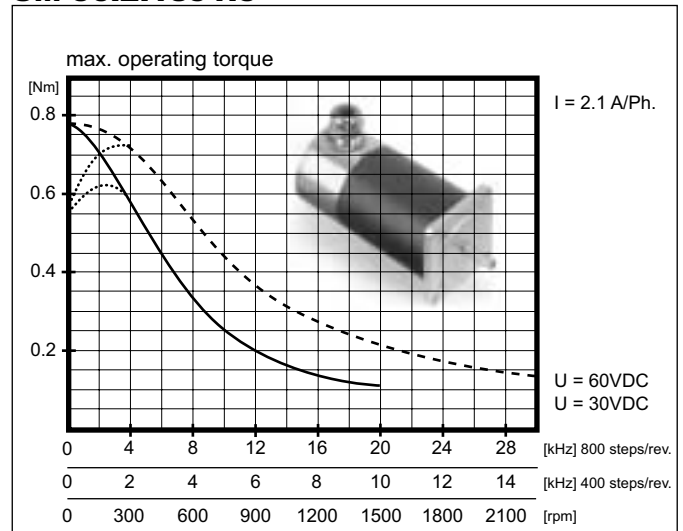
All printed torque performance curves are measured with STÖGRA drives.

————— measured with 30 VDC
 - - - - - measured with 60 VDC
 without boost

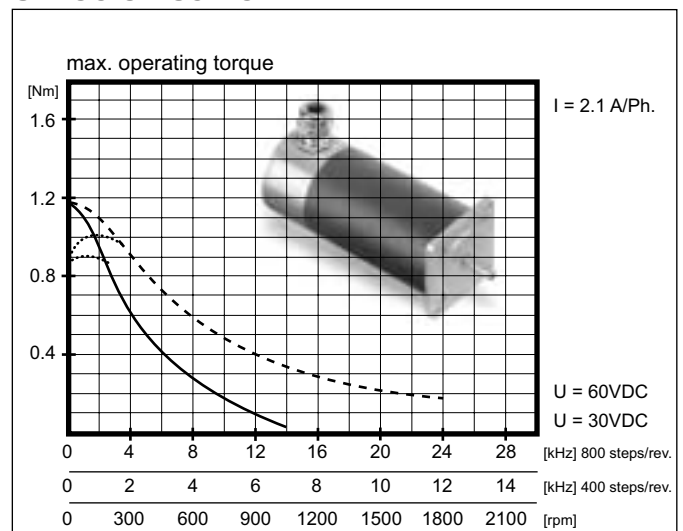
SM 56.1.18J1



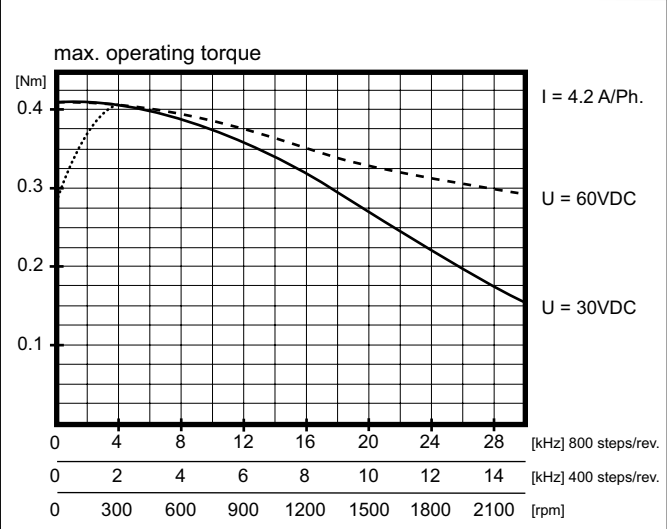
SM 56.2.18J1.5



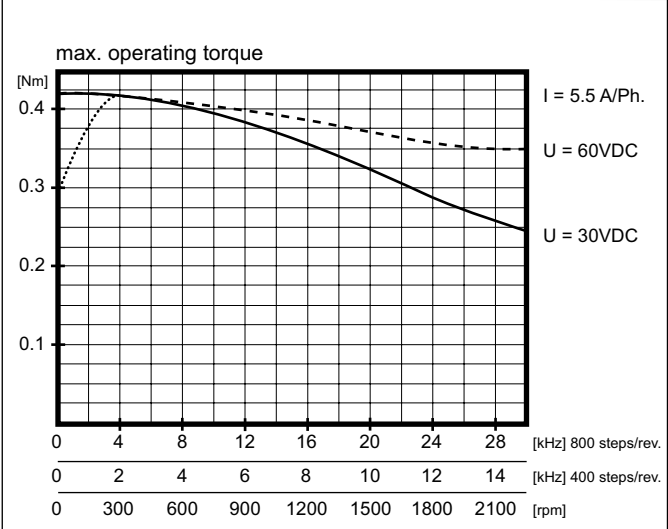
SM 56.3.18J1.5



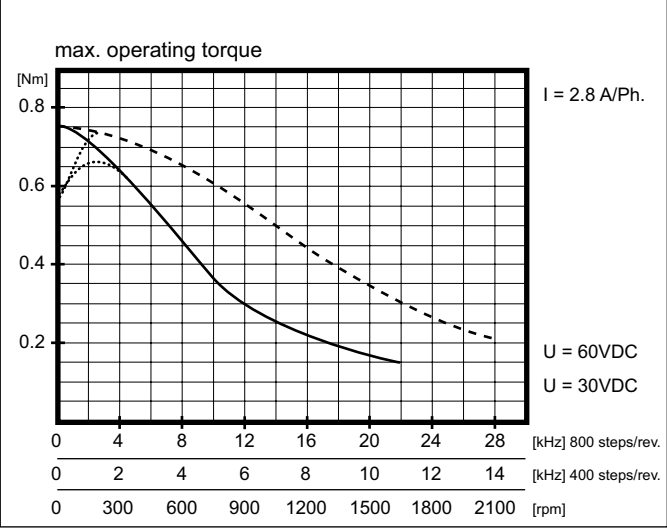
SM 56.1.18J3



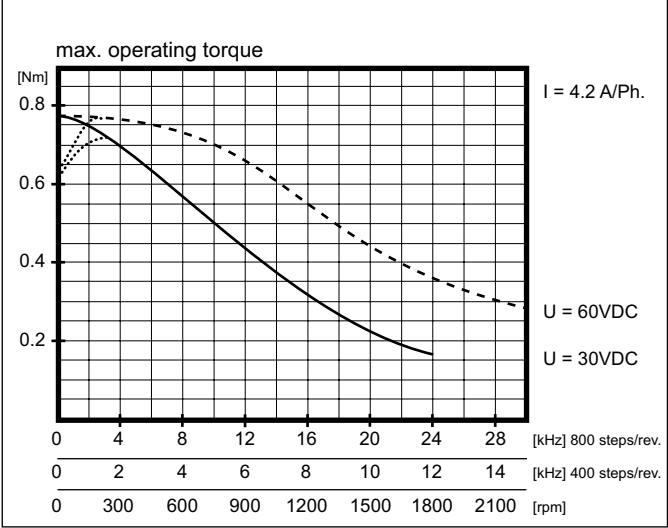
SM 56.1.18J3.9



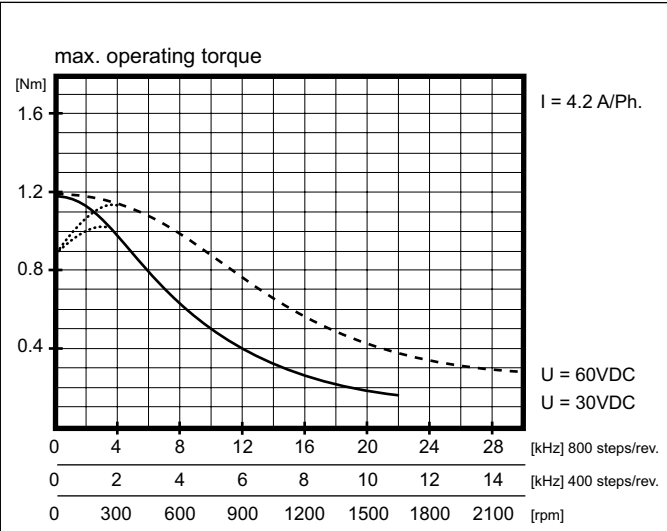
SM 56.2.18J2



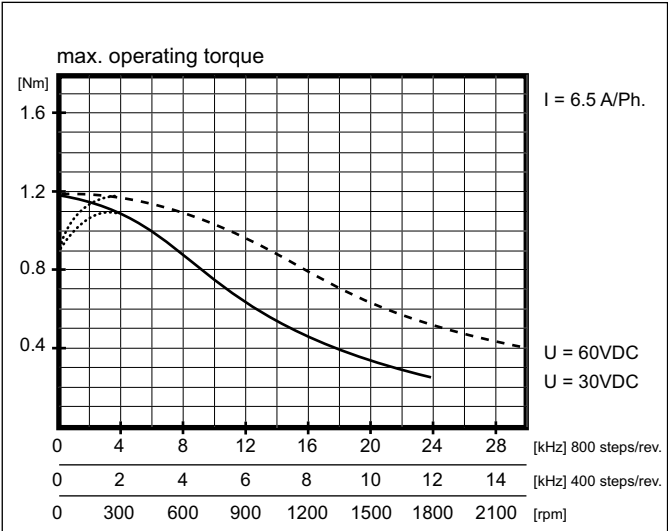
SM 56.2.18J3



SM 56.3.18J3



SM 56.3.18J4.6



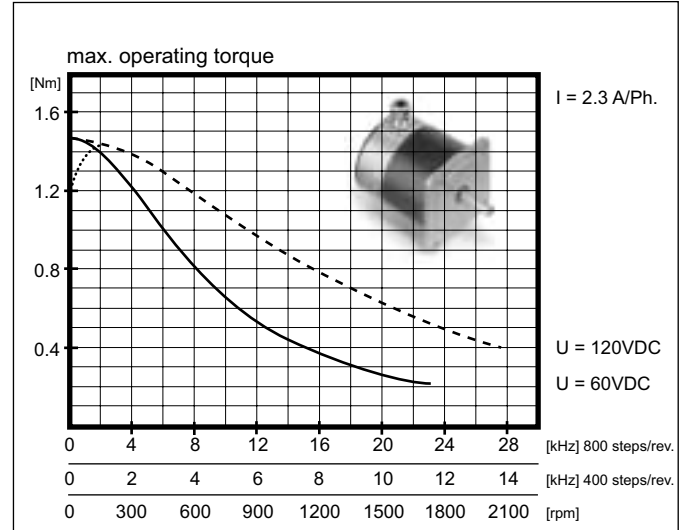
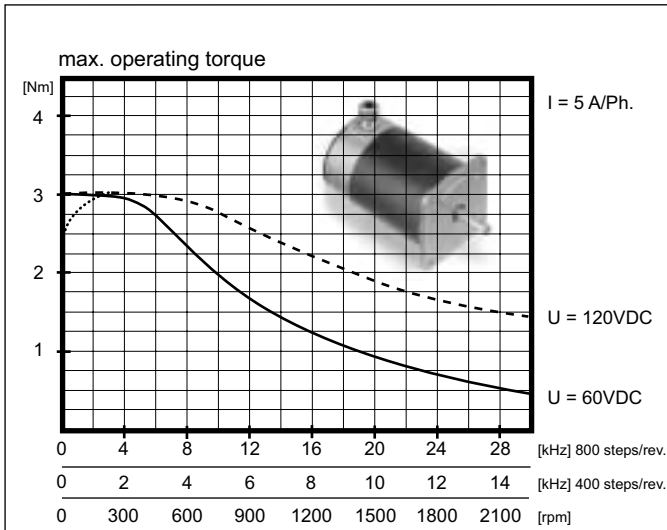
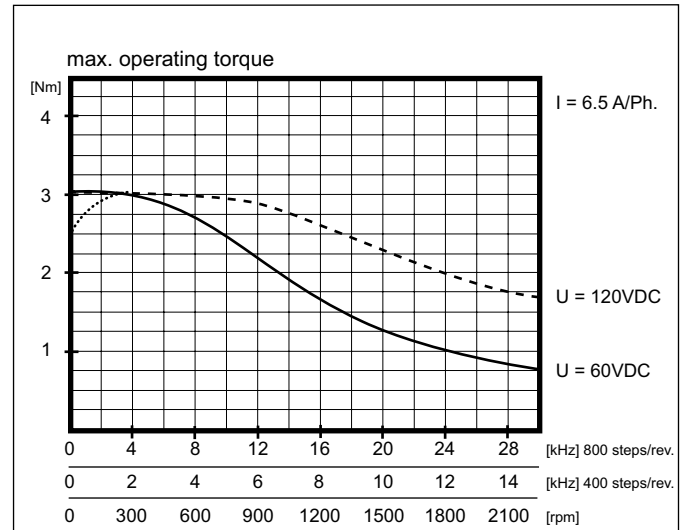
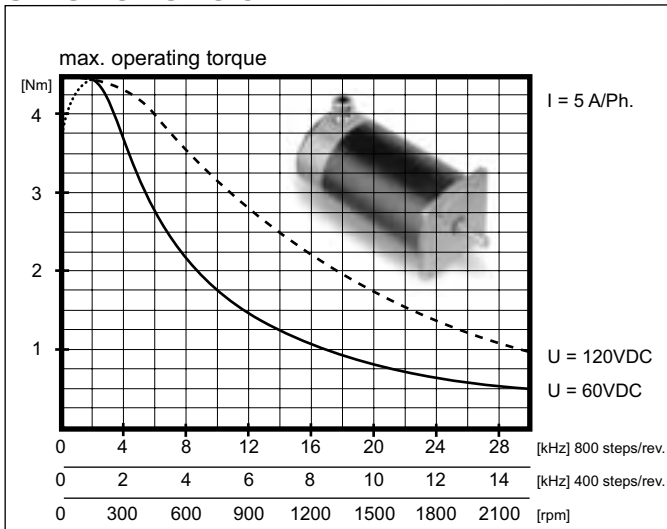
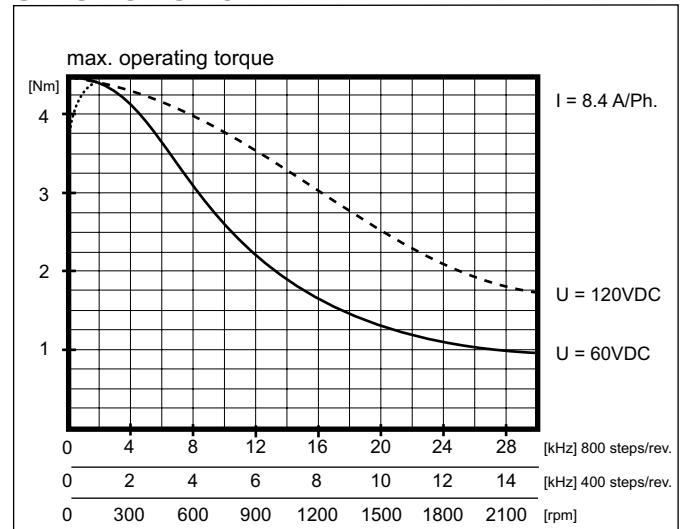
10 max. operating torque series SM 87



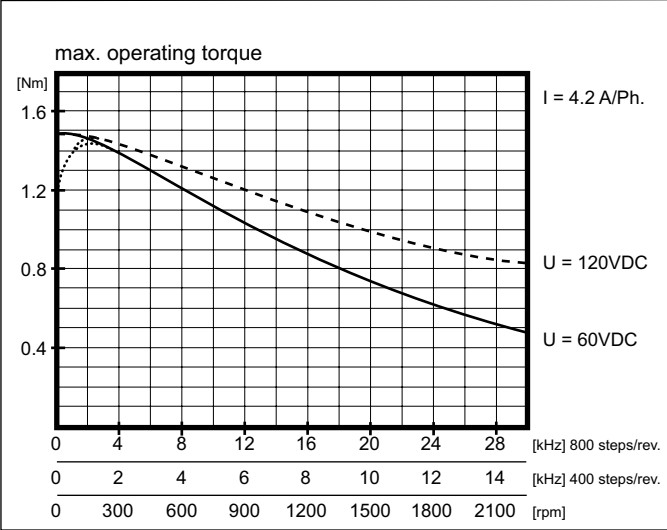
SM 168 SM 107 SM 87/88 SM 56

All printed torque performance curves are measured with STÖGRA drives.

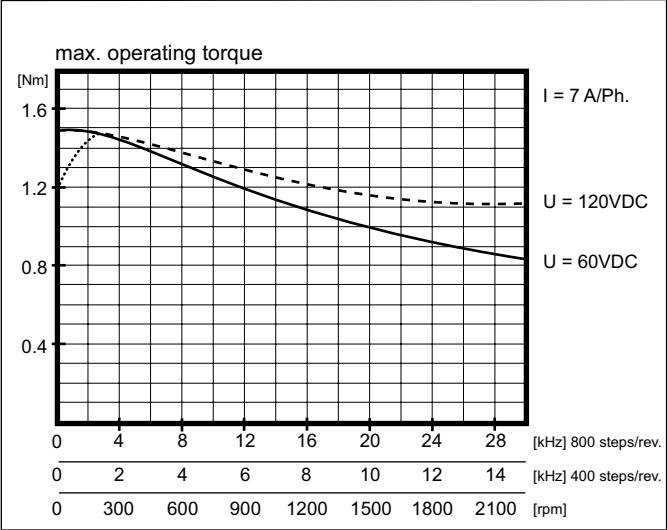
————— measured with 60 VDC
 - - - - - measured with 120 VDC
 without boost

SM 87.1.18M1.6**SM 87.2.18M3.5****SM 87.2.18M4.6****SM 87.3.18M3.5****SM 87.3.18M6**

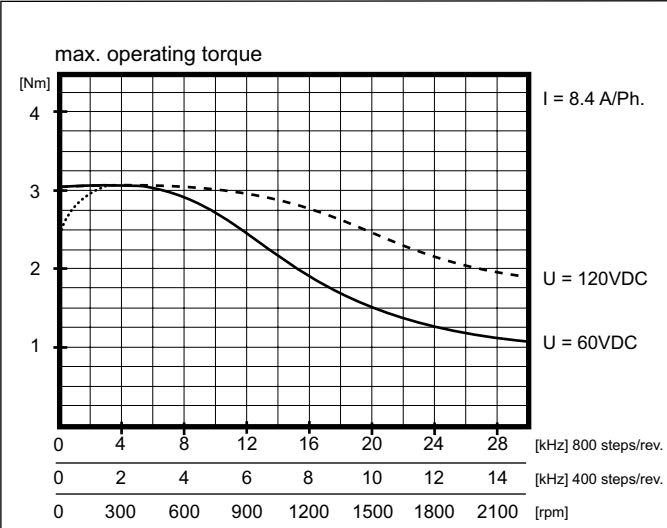
SM 87.1.18M3



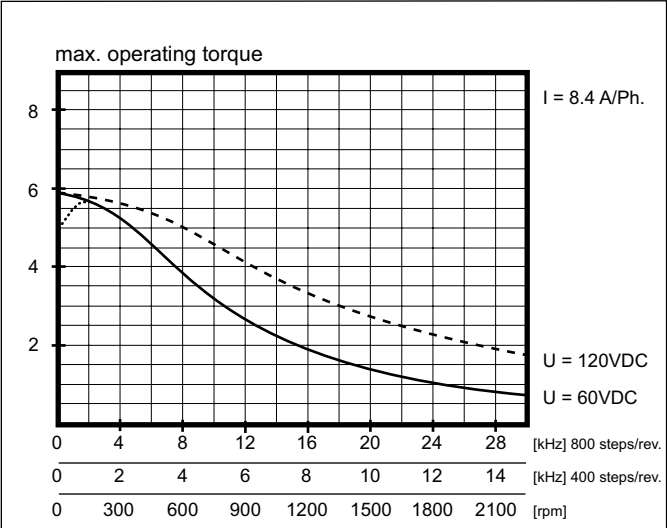
SM 87.1.18M5



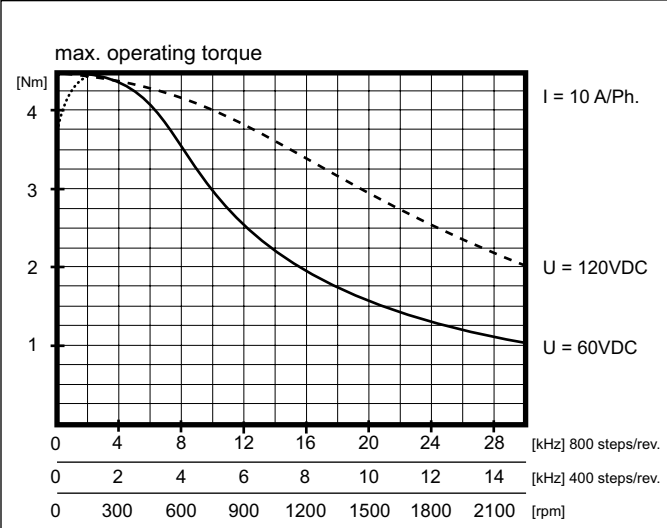
SM 87.2.18M6



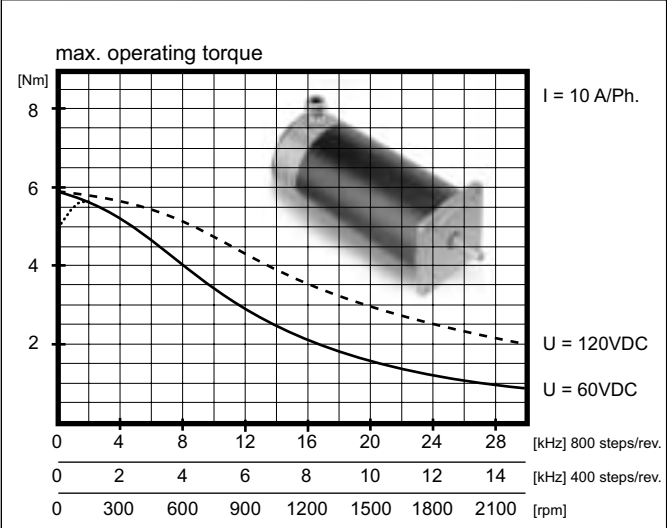
SM 87.4.18M6



SM 87.3.18M7



SM 87.4.18M7



12 max. operating torque series SM 88

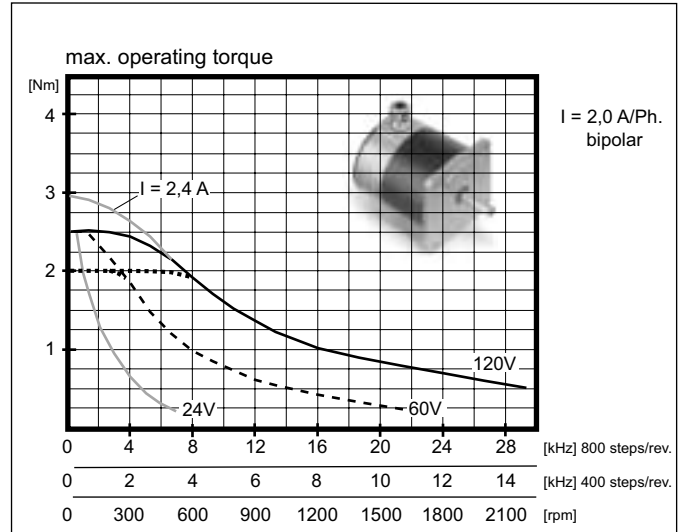


SM 168 SM 107 SM 87/88 SM 56

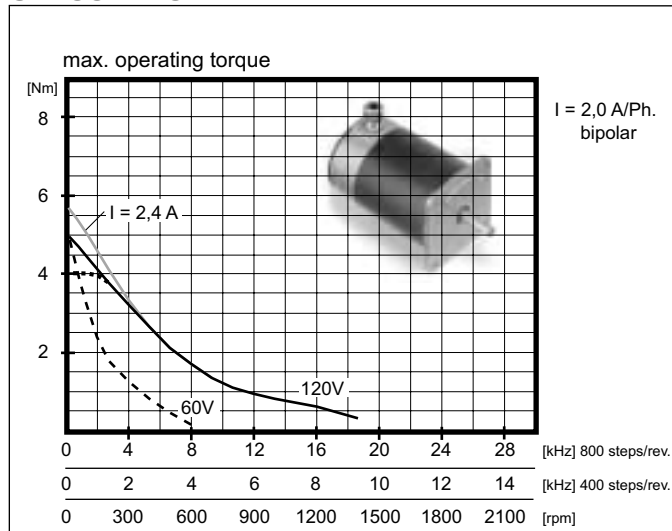
All printed torque performance curves are measured with STÖGRA drives.

- rectangular current characteristics
- rectangular current characteristics
- sinus current characteristics

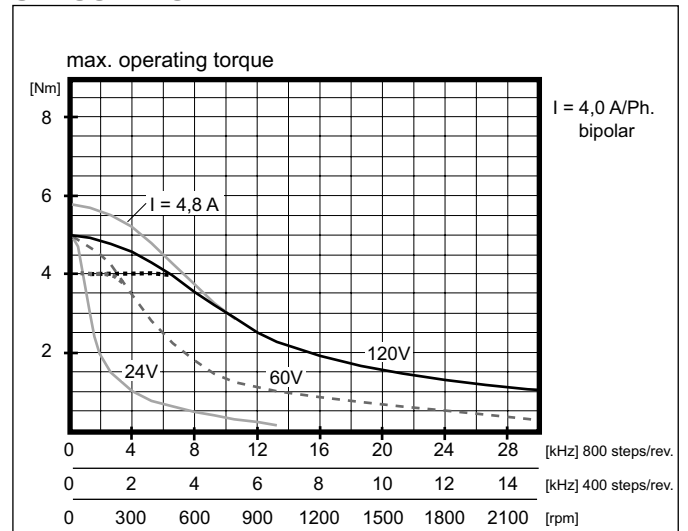
SM 88.1.18M2



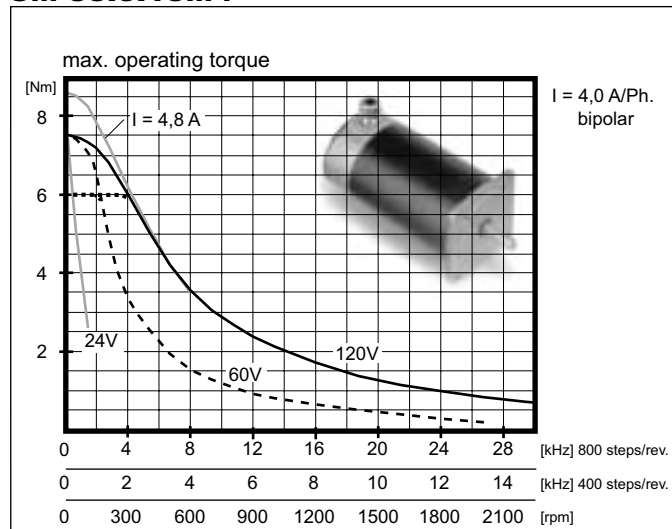
SM 88.2.18M2



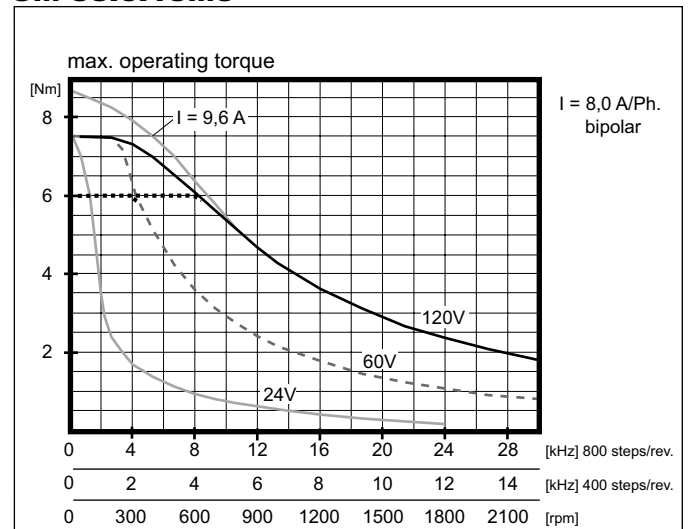
SM 88.2.18M4



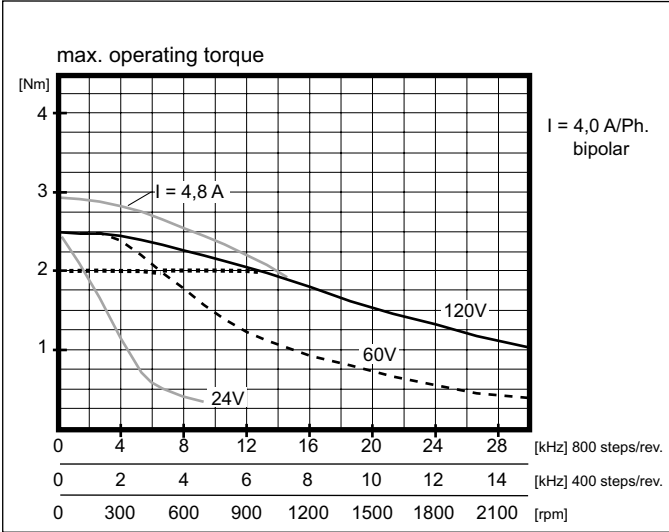
SM 88.3.18M4



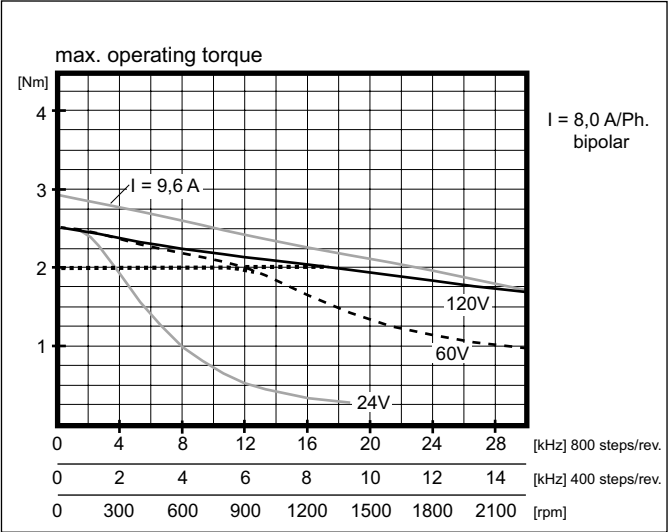
SM 88.3.18M8



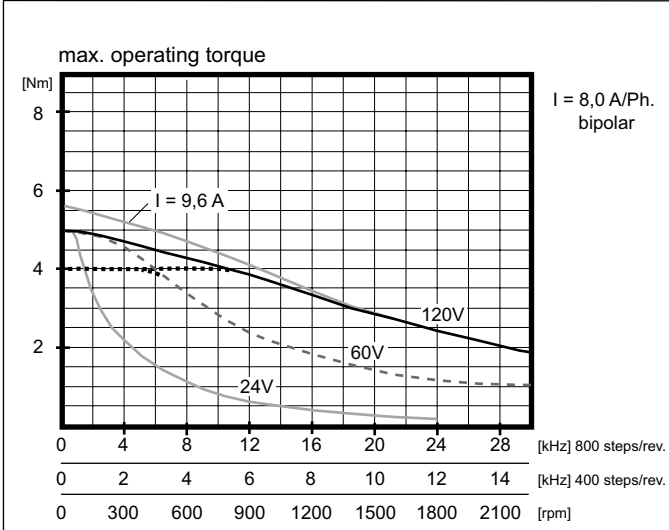
SM 88.1.18M4



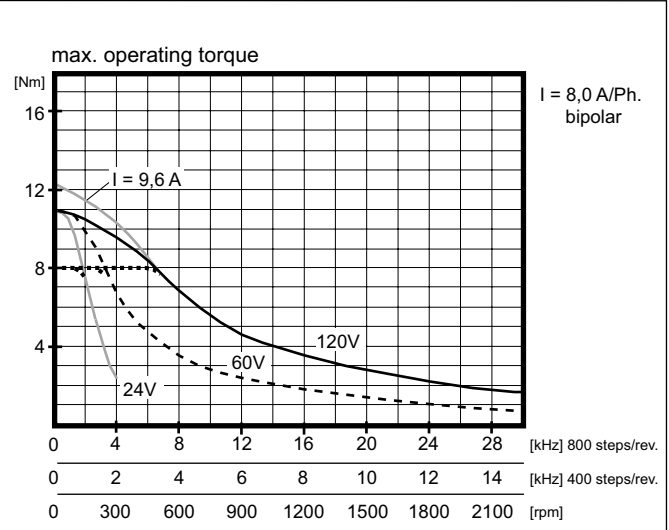
SM 88.1.18M8



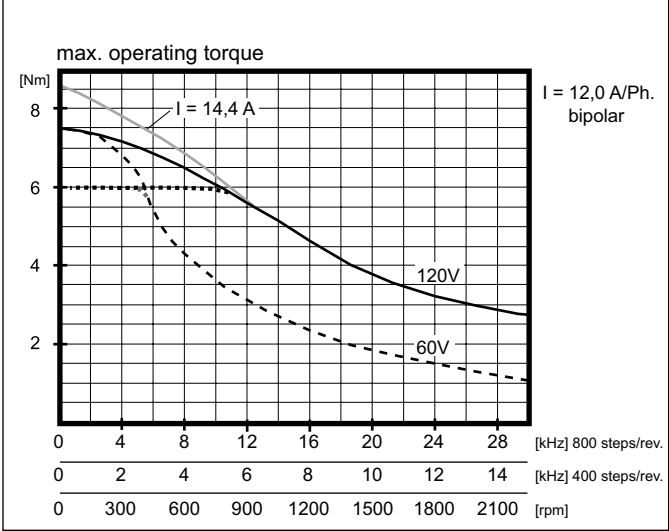
SM 88.2.18M8



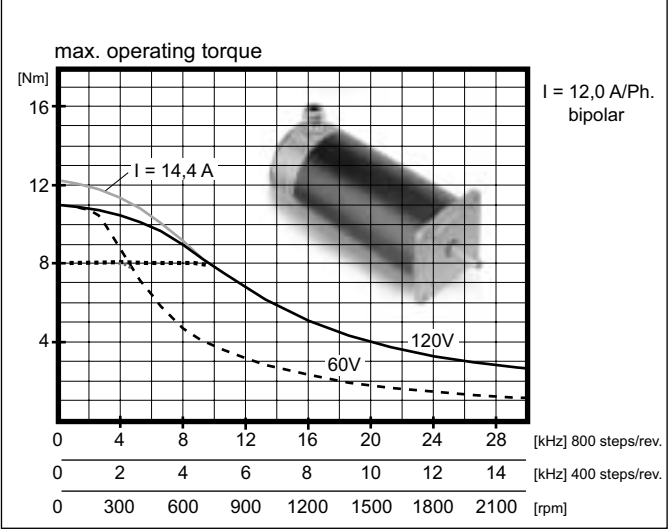
SM 88.4.18M8



SM 88.3.18M12



SM 88.4.18M12



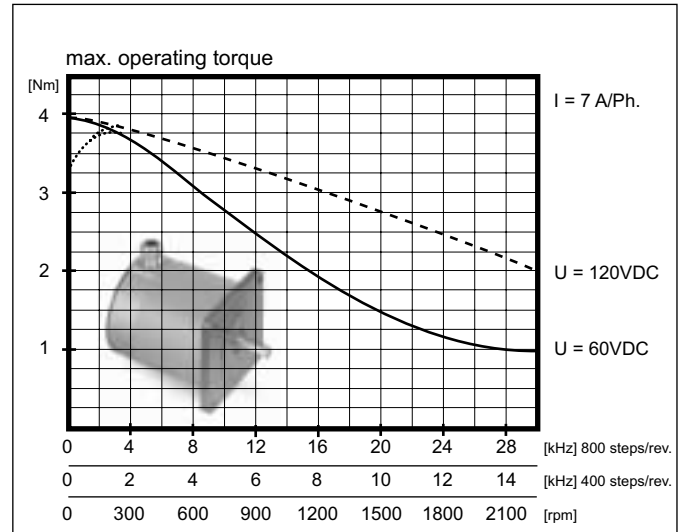
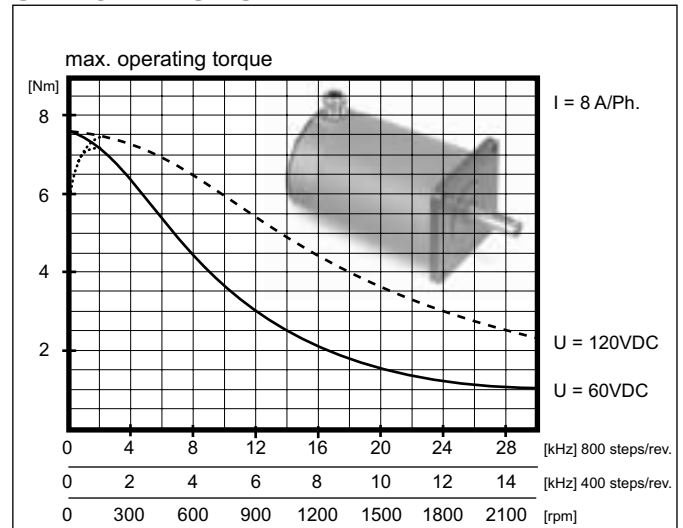
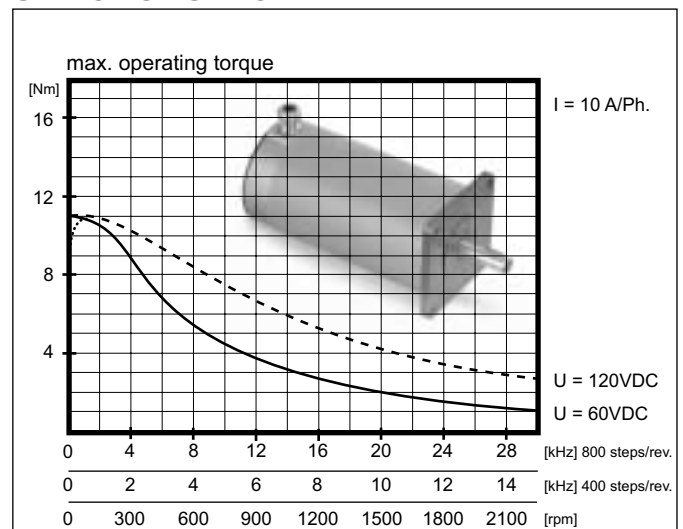
14 max. operating torque series SM 107



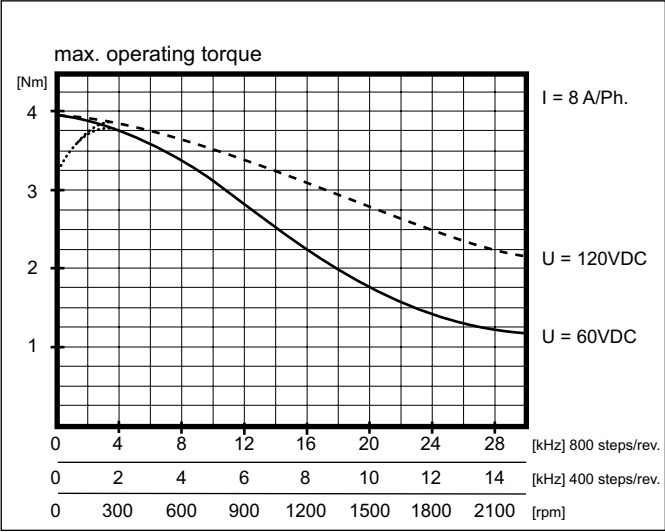
SM 168 SM 107 SM 87/88 SM 56

All printed torque performance curves are measured with STÖGRA drives.

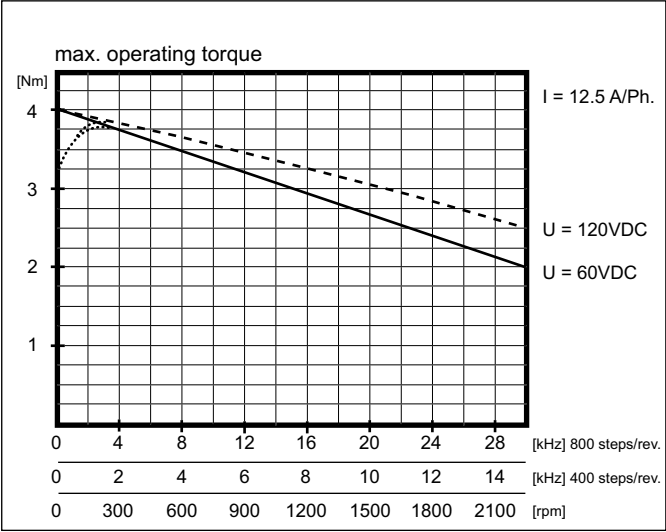
— measured with 60 VDC
 - - - measured with 120 VDC
 without boost

SM 107.1.18M6**SM 107.2.18M8****SM 107.3.18M10**

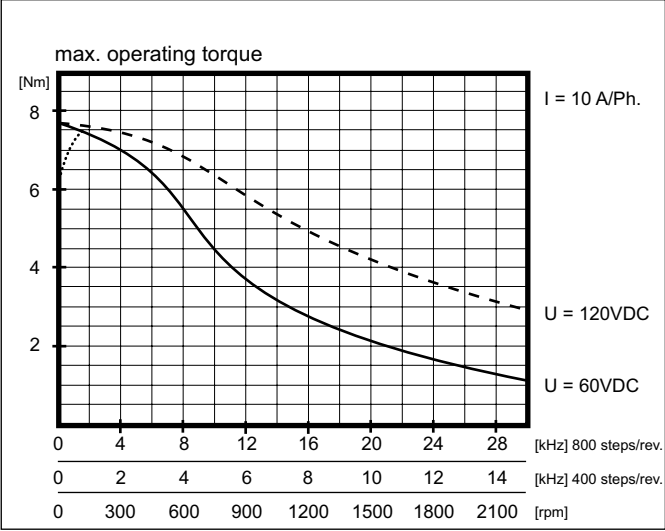
SM 107.1.18



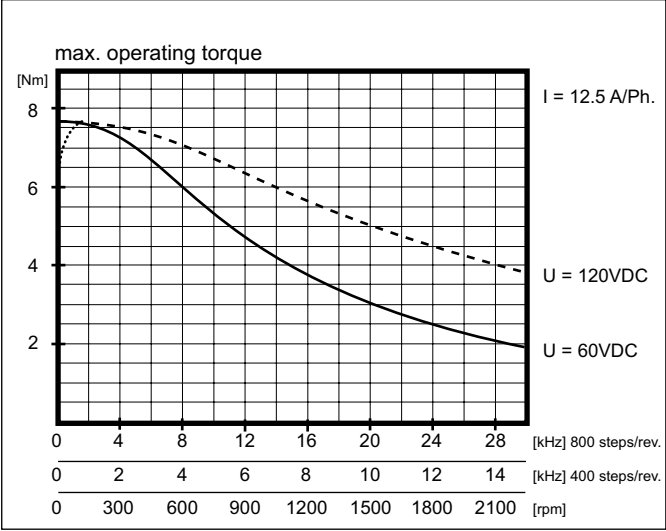
SM 107.1.18M12



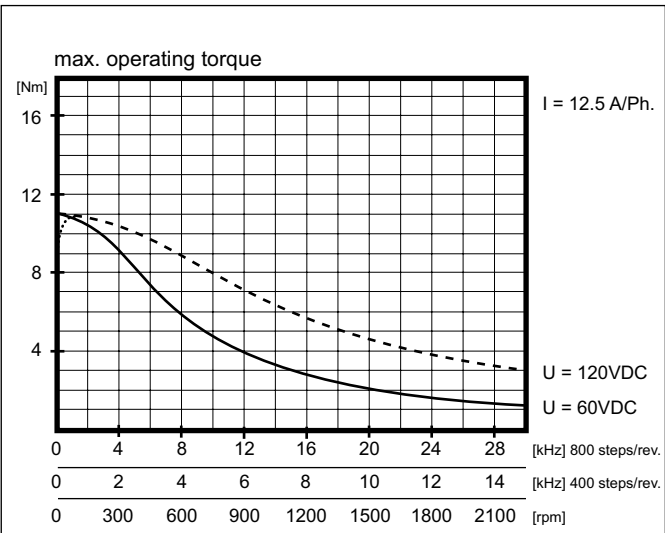
SM 107.2.18M10



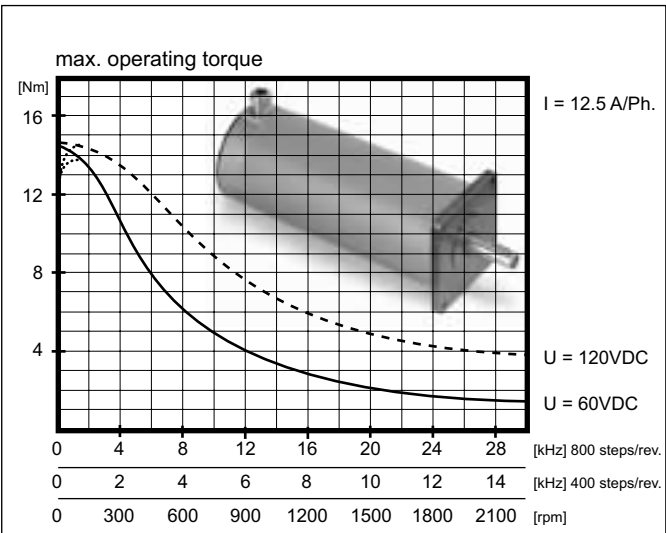
SM 107.2.18M12



SM 107.3.18M12



SM 107.4.18M12



16 max. operating torque series SM 168

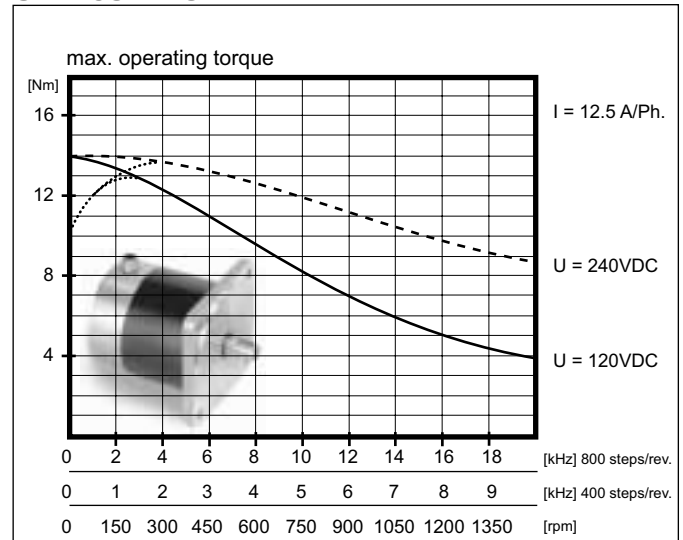


SM 168 SM 107 SM 87/88 SM 56

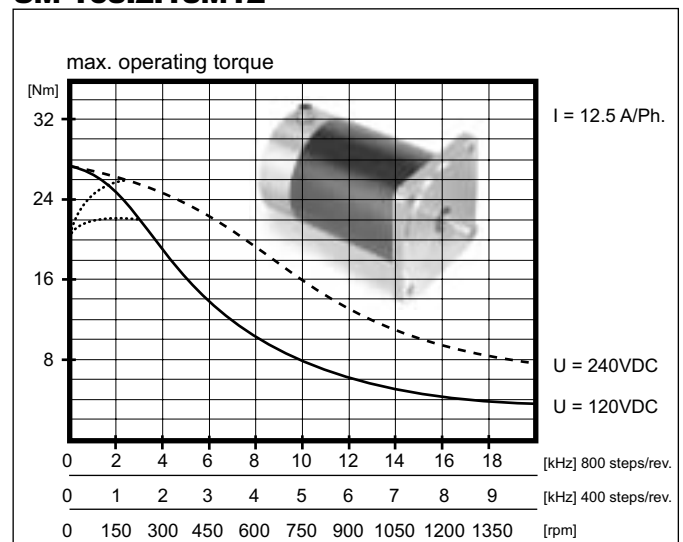
All printed torque performance curves are measured with STÖGRA drives.

————— measured with 120 VDC
 - - - - - measured with 240 VDC
 without boost

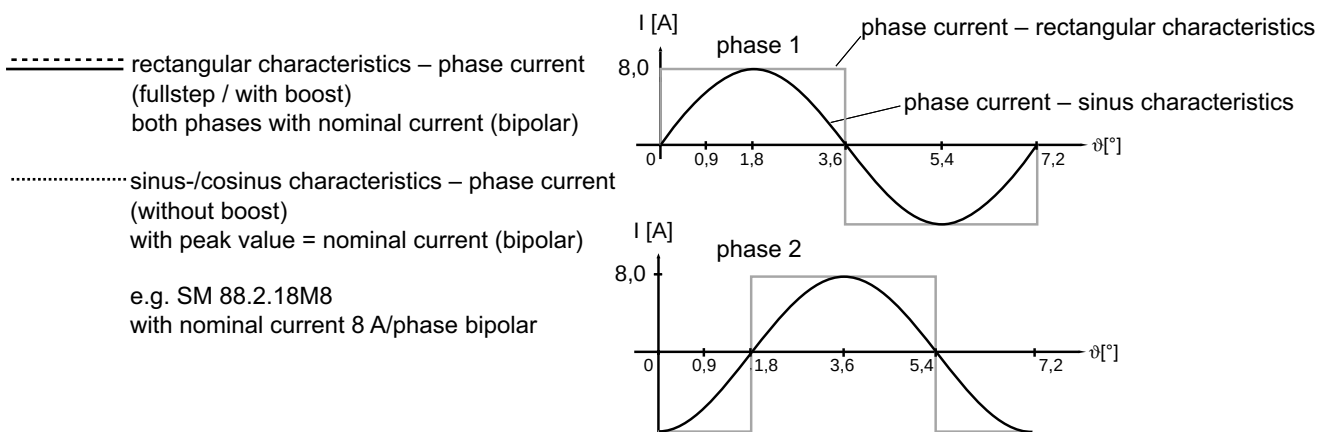
SM 168.1.18M12



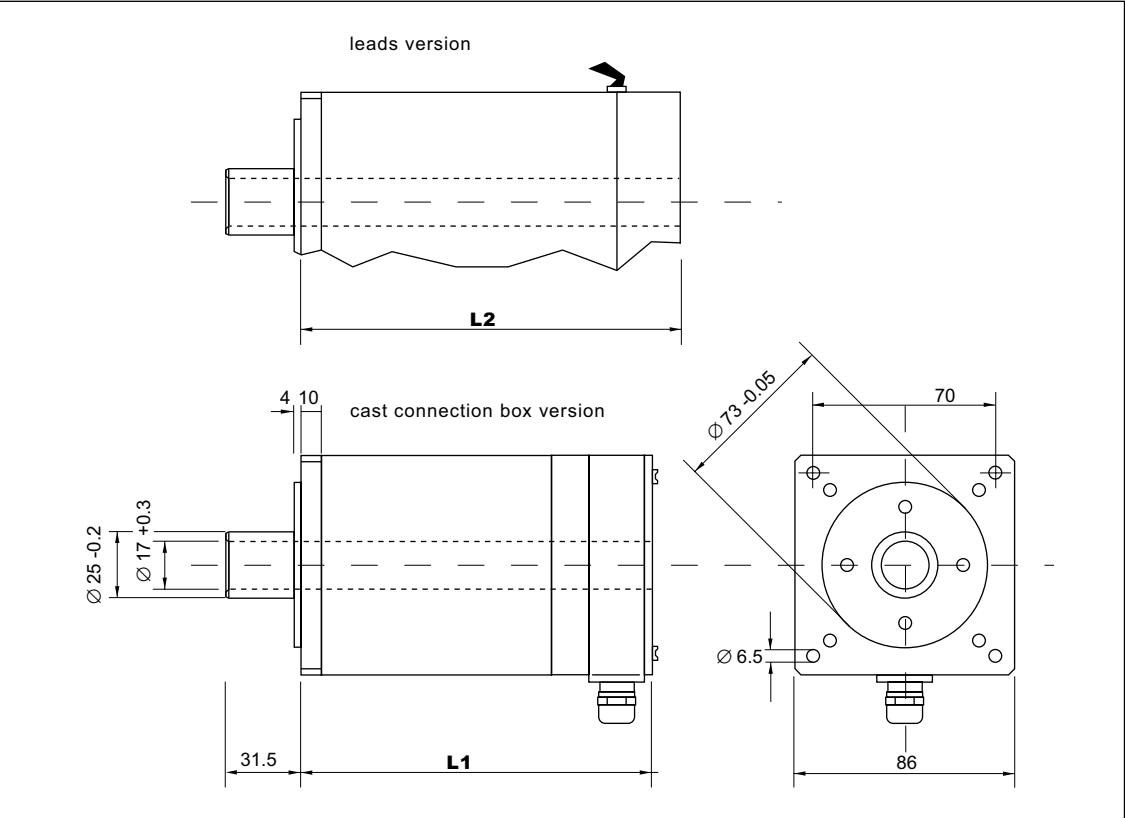
SM 168.2.18M12



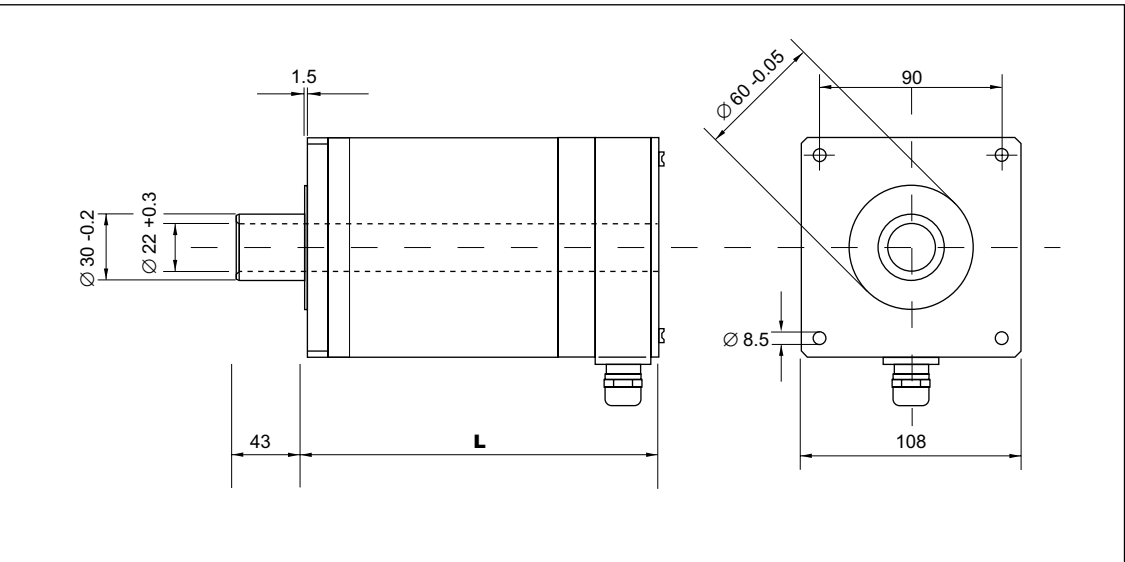
stepping motor phase current characteristics



series SMH 88



series SMH 107



series	type	L1 [mm]	L2 [mm]
SMH 88	SMH 88.1.18M ...	112.5	87
	SMH 88.2.18M ...	144.5	119
	SMH 88.3.18M ...	176.5	151
	SMH 88.4.18M ...	208.5	183

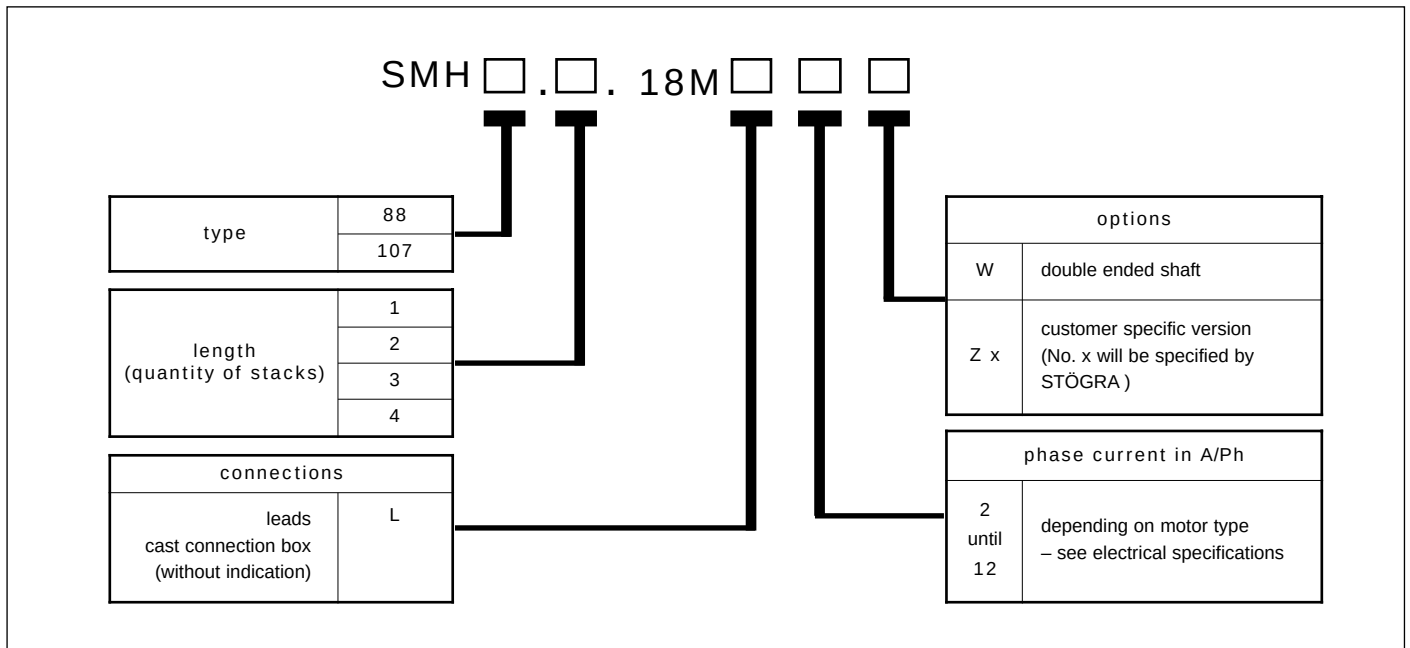
all dimensions in mm

series	type	L+1
SMH 107	SMH 107.1.18M ...	140
	SMH 107.2.18M ...	190
	SMH 107.3.18M ...	240
	SMH 107.4.18M ...	290

all dimensions in mm

18 series SM hollow shaft

type key

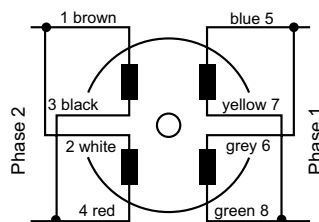


examples for ordering key: **SMH 88.2.18ML4**, **SMH 107.3.18M12**

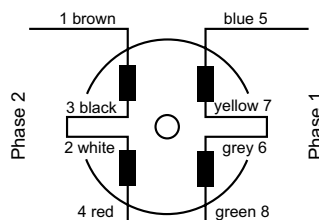
Electrical and mechanical specifications and max. operating torques for series SMH 88 are identical to series SM 88 (see page 7 and pages 12 to 13). For series SMH 107 the specifications are identical to series SM 107 (see page 7 and pages 14 to 15).

Standard stepping motor connections series SM56 – SM168

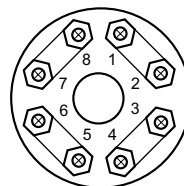
standard connections:
windings
bipolar-parallel



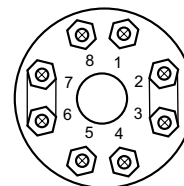
windings
bipolar-serial
(not for
series SM88)



cast connection box
version:



diagrams are valid only
for standard types
(e.g. not valid
for type SM...E50)



leads version:

standard: 8 leads
length of leads approx.
300mm

- 1 – brown
- 2 – white
- 3 – black
- 4 – red
- 5 – blue
- 6 – grey
- 7 – yellow
- 8 – green
- 9 – yellow/green: PE

series SM 88 only with
5 leads for connection
bipolar-parallel:
Ph1: blue and yellow
Ph2: brown and black
PE: yellow and green

SE z.i.u ... e.g. SE 800.06.85 V33 SERS i.u ... e.g. SERS 06.85 V03 E50
 z i u i u

u: voltage supply 24 VDC until 240 VDC

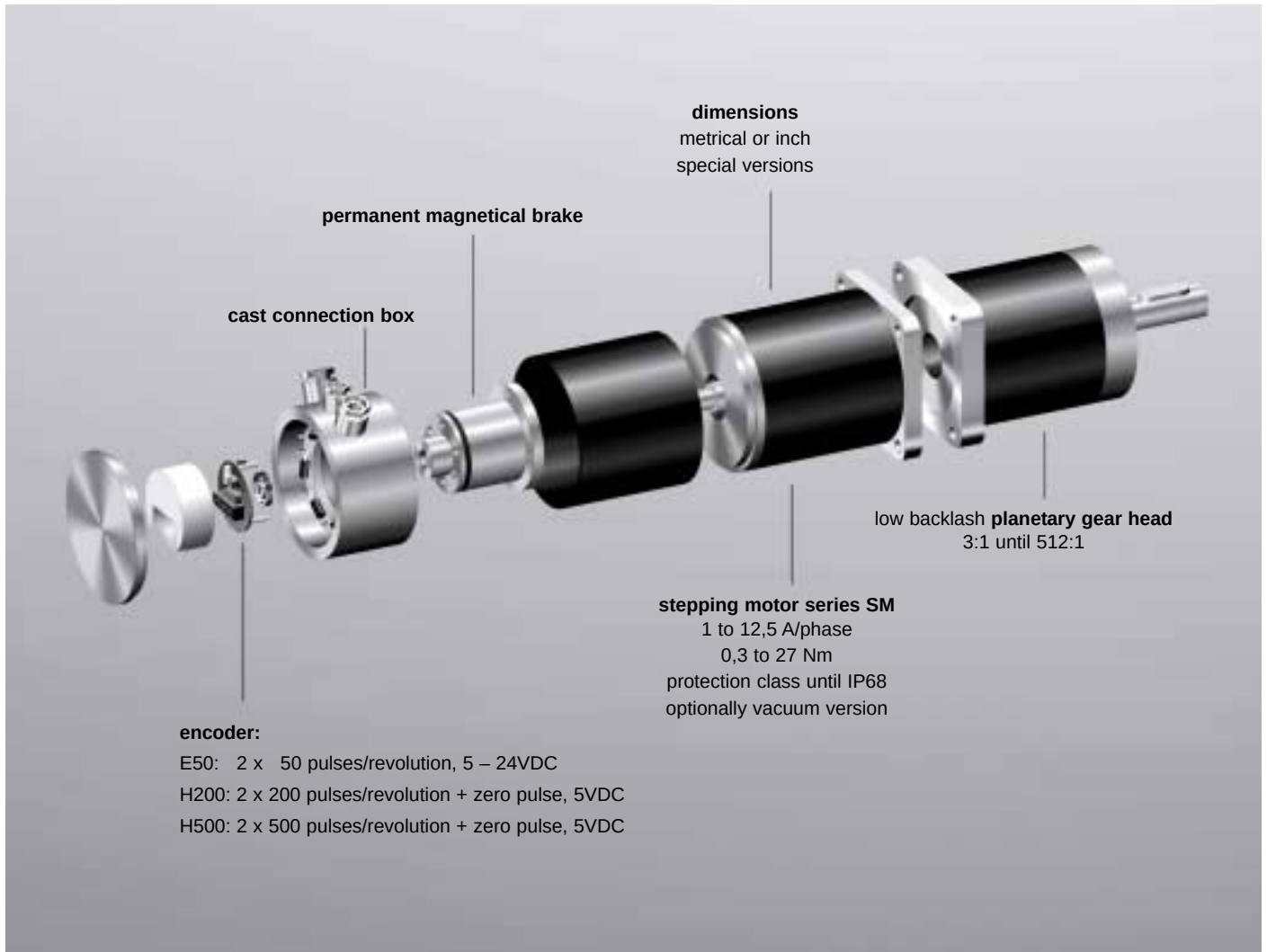
		□ = possible combinations																	
		SE z.02.24 / SERS 02.24	SE z.02.60 / SERS 02.60	SE z.03.24 / SERS 03.24	SE z.03.60 / SERS 03.60	SE z.04.24 / SERS 06.24	SE z.04.85 / SERS 06.85	SE z.04.120 / SERS 06.120	SE z.06.24 / SERS 06.24	SE z.06.85 / SERS 06.85	SE z.06.120 / SERS 06.120	SE z.08.85 / SERS 12.85	SE z.08.120 / SERS 12.120	SE z.12.85 / SERS 12.85	SE z.12.120 / SERS 12.120	SE z.12.240 / SERS 12.240	WSE 04.80	WSE 08.80	WSE 04.230 AC
series 56	SM 56.1.18 J1	□	□														□		
	SM 56.1.18 J3			□	□		□										□		
	SM 56.1.18 J3.9					□	□											□	
	SM 56.2.18 J1.5	□	□														□		
	SM 56.2.18 J2	□	□														□		
	SM 56.2.18 J3			□	□		□										□		
	SM 56.3.18 J1.5	□	□														□		
	SM 56.3.18 J3			□	□		□										□		
SM 56.3.18 J4.6								□	□	□							□		
series 87	SM 87.1.18 M1.6	□	□														□		
	SM 87.1.18 M3			□	□		□										□		
	SM 87.1.18 M5								□	□	□							□	
	SM 87.2.18 M3.5					□	□	□										□	
	SM 87.2.18 M4.6								□	□	□							□	
	SM 87.2.18 M6								□	□	□							□	
	SM 87.3.18 M3.5					□	□	□										□	
	SM 87.3.18 M6								□	□	□							□	
	SM 87.3.18 M7											□	□					□	
	SM 87.4.18 M6								□	□	□							□	
SM 87.4.18 M7											□	□					□		
series 88	SM 88.x.18 M2 ¹⁾	□	□				□	□									□		□
	SM 88.x.18 M4 ¹⁾			□	□		□	□									□		□
	SM 88.x.18 M8 ¹⁾								□	□	□							□	
	SM 88.3.18 M12												□	□					
	SM 88.4.18 M12												□	□					
series 107	SM107.1.18 M6								□	□	□							□	
	SM107.1.18 M8											□	□					□	
	SM107.1.18 M12												□	□					
	SM107.2.18 M8											□	□					□	
	SM107.2.18 M10											□	□						
	SM107.2.18 M12												□		□				
	SM107.3.18 M10											□	□						
	SM107.3.18 M12												□	□					
SM107.4.18 M12												□	□	□					
series 168	SM 168.1.18 M12													□	□	□			
	SM 168.2.18 M12													□	□	□			

¹⁾ SM 88.x stands for SM 88.1, SM 88.2, SM 88.3 and SM 88.4

20 overview: stepping motor with gear, encoder and brake

Stepping motor with gear, encoder and brake

STÖGRA-stepping motors are available with various equipment:



stepping motor series SM 56 with D-Sub connector

For series SM 56 there is a special version with connections for motor, encoder and brake via 9-pole D-Sub male and female connectors.

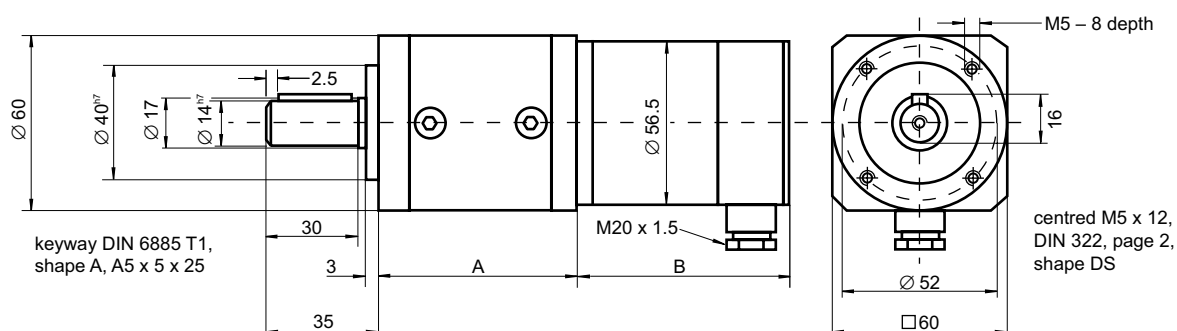


SM 56...Z154
with D-Sub connector



SM 56...E50 Z159
with D-Sub connector

degree of protection:	IP54	motor type	²⁾ length B ± 1 mm	inertia kgcm ²
torque shaft bearing:	ball bearing			
max. axial load:	600 N based on the center of the output shaft			
max. radial load:	500 N			
lubrication:	life time lubrication	SM 56.1.18	76	0.125
operating temperature:	-25°C/+90°C	SM 56.2.18	102	0.25
mounting position:	any	SM 56.3.18	130	0.375



planetary gear head series PE											
type	ratio	max. torque output Nm			inertia referred to motor shaft 10 ⁻² kgcm ²	stiffness of distortion Nm/arcmin	absolute backlash angular minutes	efficiency approx. %	length A mm	weight without motor kg	number of stages
PE3	3	1.1	2	3.2	6.5	2.3	<20	96	55	0.65	1
¹ PE4	4	1.5	2.7	4.3	3.3						
PE5	5	1.9	3.8	5.4	2.2						
¹ PE8	8	3.1	5.4	8.6	1.2						
PE9	9	3.3	5.7	9.2	7.2	2.5	<25	94	67	0.82	2
PE12	12	4.4	7.7	12.2	7						
PE15	15	5.5	9.6	15.3	2.4						
PE16	16	5.7	10.2	16.3	3.4						
PE20	20	7.3	12.8	20.4	2.4						
PE25	25	9.1	15.9	25.5	2.3						
PE32	32	11.7	20.4	32.6	1.2						
¹ PE40	40	14.6	25.5	40	1.2						
PE64	64	18	18	18	1.0						
PE60	60	20.6	36	44	2.4						
PE80	80	27,5	44	44	2.4						
PE100	100	34.4	44	44	2.4						
PE120	120	41.3	44	44	1.2						
PE160	160	44	44	44	0.1						
PE200	200	40	40	40	0.1						
PE256	256	44	44	44	0.1						
PE320	320	40	40	40	0.1						
PE512	512	18	18	18	0.1						

¹⁾ preferential versions (on stock)

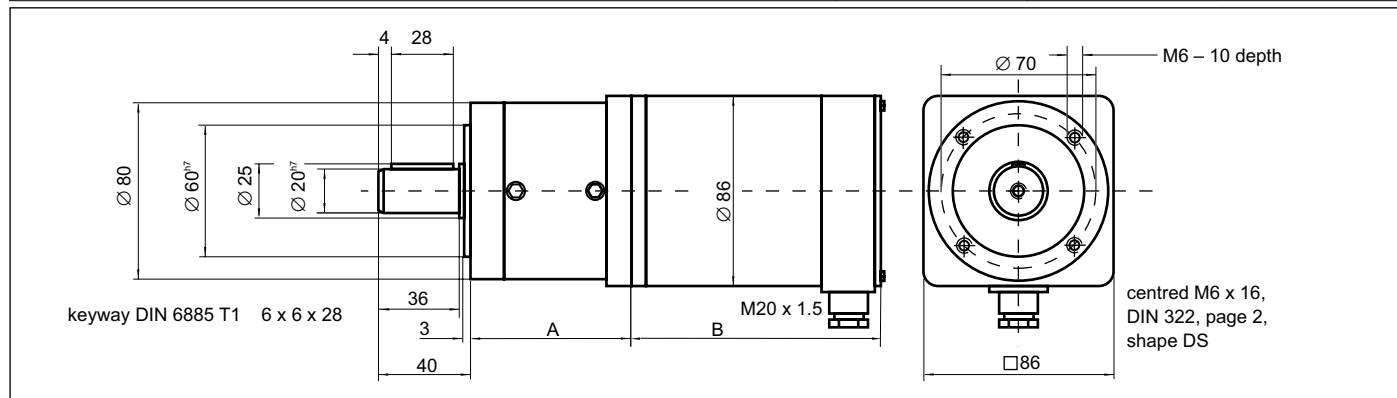
²⁾ standard motor with cast connection box

ordering indication: (example) SM 56.2.18J3 PE8

22 Stepping motor with planetary gear head series SM 87 PE / SM 88 PE

degree of protection: IP54	motor type	^{a)} length B ± 1 mm	inertia kgcm ²
torque of shaft bearing: ball bearing	SM 87.1.18	85.5	0.65
max. axial load: 1200 N based on the center of the output shaft	SM 87.2.18	117.5	1.3
max. radial load: 950 N	SM 87.3.18	149.5	1.95
lubrication: life time lubrication	SM 87.4.18	181.5	2.6
operating temperature: -25°C/+90°C	SM 88.1.18	93.5	1.35
mounting position: any	SM 88.2.18	125.5	2.7
	SM 88.3.18	157.5	4.05
	SM 88.4.18	189.5	5.4

without gear head



planetary gear head series PE																
type	ratio	max. torque output Nm								inertia referred to motor shaft 10 ⁻² kgcm ²	stiffness of torsion Nm/arcmin	absolute backlash angular-minutes	efficiency approx. %	length A mm	weight without motor kg	number of stages
PE3	3	4.3	7.2	8.6	14.4	12.9	21.6	17.3	28.8	63	6	<12	96	72	1.6	1
¹⁾ PE4	4	5.8	9.6	11.5	19.2	17.2	28.8	23	38.4	25						
PE5	5	7.2	12	14.4	24	21.6	36	28.8	48	14						
¹⁾ PE8	8	11.5	19.2	23	38.4	34.5	50	46.1	50	8						
PE9	9	12.7	21.1	25.4	42.3	38	63.4	50.7	84.6	63	6.5	<17	94	89	2.2	2
PE12	12	16.9	28.2	33.8	56.4	50.7	84.6	67.7	112.8	26						
PE15	15	21.2	35.2	42.3	70.5	63.4	105.8	84.6	110	62						
PE16	16	22.6	37.6	45.1	75.2	67.6	112.8	90.2	120	25						
PE20	20	28.2	47	56.4	94	84.6	120	112.8	120	15						
PE25	25	35.3	58.7	70.5	110	105.7	110	110	110	15						
PE32	32	45.1	75.2	90.2	120	120	120	120	120	8						
¹⁾ PE40	40	56.4	94	110	110	110	110	110	110	8						
PE64	64	50	50	50	50	50	50	50	50	6	6.3	<22	90	106.5	2.8	3
PE60	60	81	110	110	110	110	110	110	110	25						
PE80	80	108	120	120	120	120	120	120	120	18						
PE100	100	120	120	120	120	120	120	120	120	15						
PE120	120	110	110	110	110	110	110	110	110	60						
PE160	160	120	120	120	120	120	120	120	120	8						
PE200	200	110	110	110	110	110	110	110	110	8						
PE256	256	120	120	120	120	120	120	120	120	8						
PE320	320	110	110	110	110	110	110	110	110	6						
PE512	512	50	50	50	50	50	50	50	50	6						

¹⁾ preferential versions (on stock)

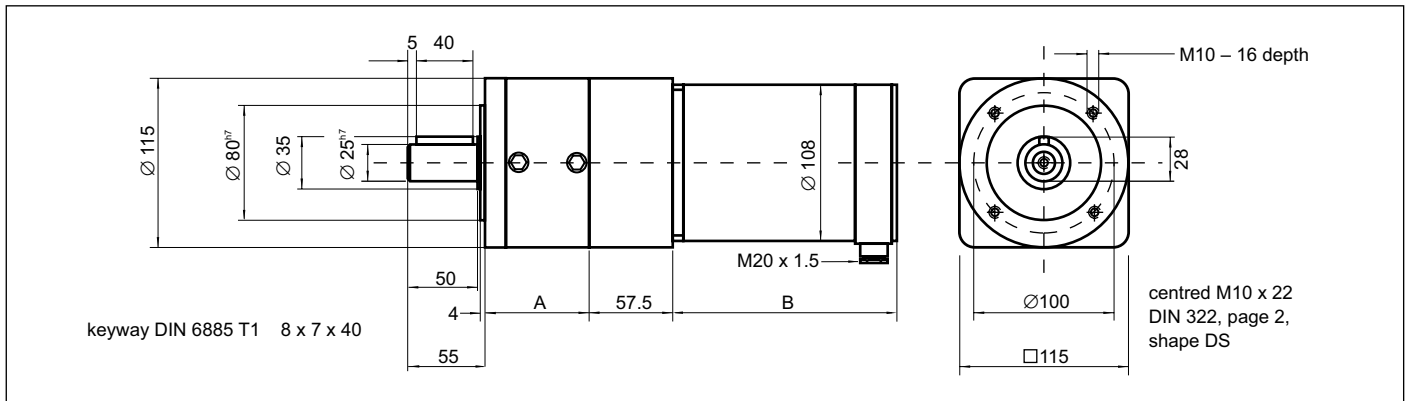
ordering indication: (example) SM 87.1.18M3 PE8

²⁾ standard motor with cast connection box

Stepping motor with planetary gear head series SM 107 PE 23

degree of protection: IP54	motor type	²⁾ length B ± 1 mm	inertia kgcm ²
torque of shaft bearing: ball bearing			
max. axial load: 2800 N based on the center of the output shaft	SM 107.1.18	111	4
max. radial load: 2000 N	SM 107.2.18	161	8
lubrication: life time lubrication	SM 107.3.18	211	12
operating temperature: -25°C/+90°C	SM 107.4.18	261	16
mounting position: any			

without gear head



planetary gear head series PE												
type	ratio	max. torque output Nm				inertia referred to motor shaft 10 ⁻² kgcm ²	stiffness of distortion Nm/arcmin	absolute backlash angular minuten	efficiency approx. %	length A mm	weight without motor kg	number of stages
PE3	3	10.8	20.8	29.7	39.4	2.6	12	<8	96	74	6.5	1
¹⁾ PE4	4	14.4	27.7	39.6	52.6	1.79						
PE5	5	18	34.7	49.5	65.7	1.53						
¹⁾ PE8	8	28.8	55.4	79.2	105	1.32						
PE9	9	30.6	58.9	84.2	112	2.62	13	<12	94	101	9	2
PE12	12	40.8	78.5	112	149	2.56						
PE15	15	51	98.2	140	186	2.53						
PE16	16	54.4	105	150	198	1.75						
PE20	20	68	131	187	248	1.5						
PE25	25	85	164	230	230	1.49						
PE32	32	109	209	260	260	1.3						
¹⁾ PE40	40	136	230	230	230	1.3						
PE64	64	120	120	120	120	1.3	12	<16	90	128	115	3
PE60	60	192	260	260	260	2.57						
PE80	80	256	260	260	260	1.5						
PE100	100	260	260	260	260	1.5						
PE120	120	230	230	230	230	2.5						
PE160	160	260	260	260	260	1.3						
PE200	200	230	230	230	230	1.3						
PE256	256	260	260	260	260	1.3						
PE320	320	230	230	230	230	1.3						
PE512	512	120	120	120	120	1.3						

¹⁾ preferential versions (on stock)

ordering indication: (example) SM 107.2.18M12 PE8

²⁾ standard motor with cast connection box

24 stepping motor – equipment

Stepping motor with integrated encoder

In non disturbed operation the stepping motor runs synchronously to the pulses coming from the control, that means the motor rotation (= rotation of the rotor) is synchronously to the pulse frequency (= rotating stator field in the motor).

In case of a load at the motor (e.g. via a static load at the motor shaft or because of accelerating the motor – dynamical load) the running of the motor will differ from the pulse frequency within a short time and within a certain max range. This results in changing the load angle (= difference between the real position of the rotor and its position command value).

Stepping motor with integrated encoder E50

The encoder series E50 monitors the movement of the motor. Together with a STÖGRA-drive series SE... E50 or SERS .. E50 the load angle of the stepping motor can be controlled. When exceeding the max. load angle allowed (e.g. in case of the motor running is interrupted because of a mechanical overload) the drive will create an error signal.



special characteristics

- simple and robust low cost version
- no changes of the motor dimensions in comparison to the standard version with cast connection box (except SM 56)
- the encoder is integrated in the motor housing. There is no additional measure necessary for protection – available until IP68
- All requirements for mechanical and climatical ambient conditions (vibration-, shock resistance, temperature and humidity) are fulfilled.
- Evaluation of the encoder signals and realisation of a step angle control with generating an error signal can be done by using standard **STÖGRA power amplifier boards series SE...E50 or series SERS...E50**

Specifications E50

electrical specifications

voltage supply: 5 – 24VDC

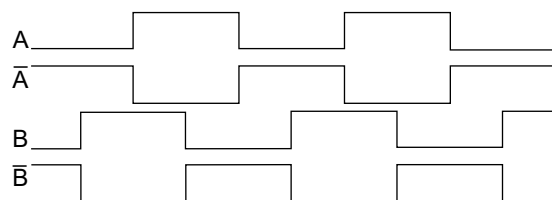
current consumption: typ. aprox 35mA (no load at outputs) – max. load at outputs 100mA / output

operating temperature: max. 100°C

outputs

- 2 x 50 pulses/revolution – signals A und B with rectangular shape and inverted signals $\bar{A}$ and $\bar{B}$
- duty cycle 1:1 $\pm$ max. 10% error
- Bipolar – switching to VCC and GND
- short circuit protected
- pulse frequency min. 20 kHz

signal outputs



connections

connection via screw terminals for nominal cross section max. 1mm² (26 – 16AWG)

for series SM56 also available with D-Sub connector (see page 20 – figures below)

dimensions as standard motor types! (except series SM56 – see table page 27)

Stepping motor with integrated encoder H200 and H500



special characteristics

- The encoder is integrated in the motor housing. There is no additional measure necessary for protection
- available until protection class IP68
- All requirements for mechanical and climatical ambient conditions (vibration-, shock resistance, temperature and humidity) are fulfilled
- Evaluation of the encoder signals and realisation of a step angle control with generation of an error signal can be done by using standard **STÖGRA** drivers SERS...E50...

Technical specifications H200 and H500

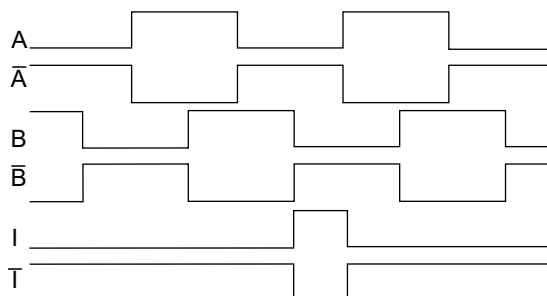
general specifications H200 and H500

- optical encoder
- voltage supply: 5 VDC
- operating temperature: max. 100°C

outputs

- 2 x 200 pulse / revolution for H200
2 x 500 pulse / revolution for H500
rectangular shape signals A und B, with inverted signals $\bar{A}$ and $\bar{B}$
- H200 and H500: zero pulse and inverted zero pulse – 1 pulse / revolution
- duty cycle 1:1 $\pm$ max. 10% error
- RS422-line driver
- short circuit protected
- pulse frequency min. 100 kHz

signal outputs



dimensions

the modified dimensions – in comparison to the standard motors without encoder – can be found in the table at page 27

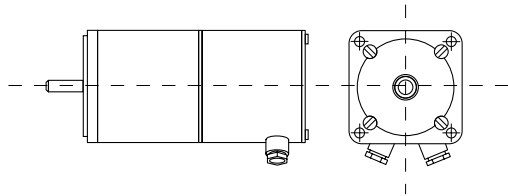
connections

connection via screw terminals for nominal cross section max. 1mm² (26 –16AWG)

26 stepping motor – equipment

Stepping motor with holding brake

brake type: **permanent magnetical brake**
 voltage supply: **24VDC**
 protection: **as motor protection – until IP68**
the brake is integrated in the motor housing



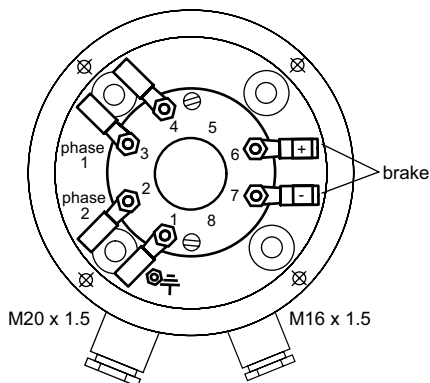
Holding brakes are used as emergency brakes in vertical axis (Z-axis). If the motor current is switched off or in case of a power line fault the motor shaft will be retarded.

The brake must be supplied by 24VDC to release the motor shaft.

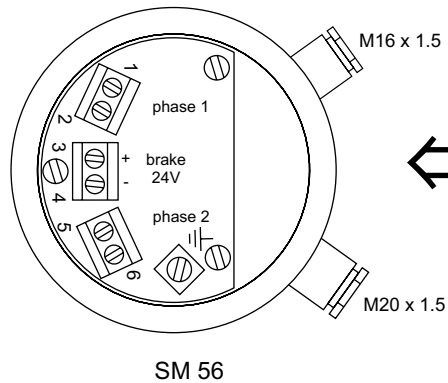
The dimensions of the different motor types with brake can be found in the table of page 27.

stepping motor with holding brake				brake				
series	type	weight kg	rotor inertia		holding torque Nm	switching time		electr. power / consumption at 24 VDC W / mA
			total kgcm ²	brake kgcm ²		ON ms	OFF ms	
56	SM 56.1.18...B	0.8	0.15	0.02	1	7	15	10 / 417
	SM 56.2.18...B	1.2	0.27					
	SM 56.3.18...B	1.6	0.4					
87	SM 87.1.18...B	2.1	0.8	0.15	4.5	20	15	12 / 500
	SM 87.2.18...B	3	1.45					
	SM 87.3.18...B	4	2.1					
	SM 87.4.18...B	5	2.75					
88	SM 88.1.18...B	2.3	1.5	0.15	4.5	20	15	12 / 500
	SM 88.2.18...B	3.2	2.85					
	SM 88.3.18...B	4.2	4.2					
	SM 88.4.18...B	5.2	5.55					
107	SM 107.1.18...B	5.2	4.65	0.63	6	35	25	13 / 542
	SM 107.2.18...B	8.1	8.65					
	SM 107.3.18...B	10.7	12.65					
	SM 107.4.18...B	13.4	16.65					

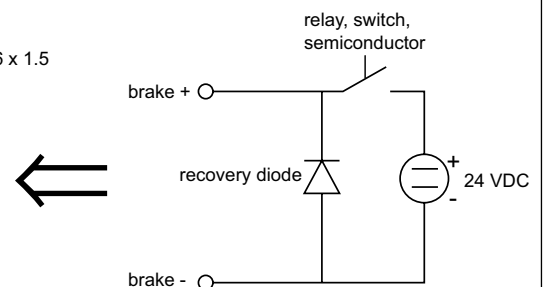
Connections brake



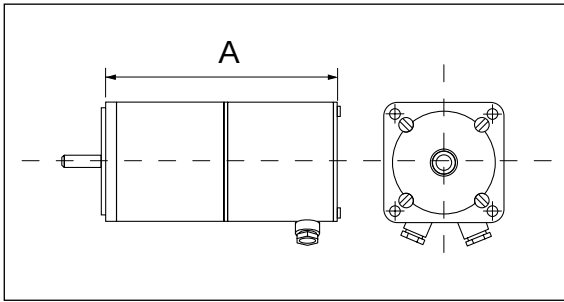
SM 87, SM 88, SM 107



SM 56



Dimensions stepping motor with gear box, encoder and brake



SM 87.2 with brake



SM 87.3 with encoder H200



SM 87.2 with brake and encoder H200

total length A ±0.5 mm	standard cast connection box	brake	E50	H200/H500	brake and E50	brake and H200/H500
SM 56.1	76	116	88	98	128	137.5
SM 56.2	102	142	114	124	154	163.5
SM 56.3	130	170	142	152	182	191.5
SM 87.1	85.5	131	85.5	104	131	153
SM 87.2	117.5	163	117.5	136	163	185
SM 87.3	149.5	195	149.5	168	195	217
SM 87.4	181.5	227	181.5	200	227	249
SM 88.1	93.5	139	93.5	112	139	161
SM 88.2	125.5	171	125.5	144	171	193
SM 88.3	157.5	203	157.5	176	203	225
SM 88.4	189.5	235	189.5	208	235	257
SM 107.1	111	168	111	136	168	193
SM 107.2	161	218	161	186	218	243
SM 107.3	211	268	211	236	268	293
SM 107.4	261	318	261	286	318	343
SM 168.1	179		179			
SM 168.2	254		254			

SM cable gland



thread	for cable Ø mm	length of thread mm	width across flats mm	connections cast connection box
M16 x 1.5	5...9	5.0	17	encoder and brake
M20 x 1.5	9...13	6.0	22	stepper motor phases

metric EMC cable glands:

- protection type IP68 up to 5 bar
- material brass nickel plated
- for shielded cables
- sealing insert made of neoprene
- O-ring on external thread



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